Department of Engineering

CSI-Feedback-Driven Link Adaptation in 5G NR Downlink MIMO: A Link-Level Simulation Study

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I hereby declare that, except where specifically indicated, the work submitted herin is my own original work.

Signed Kaiyuan Bao 包铠源 *date* 30 May 2026

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MIMO Techniques in 5/6G (F-js851-3*)

Technical Abstract

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Background and Motivation

Channel-state-information (CSI) feedback is essential for downlink multiple-input multiple-output (MIMO) transmission in 5G New Radio (NR). Based on channel measurements at the user equipment (UE), the reported rank indicator (RI), precoding matrix indicator (PMI), and channel quality indicator (CQI) allow the base station to adapt the number of spatial layers, precoding matrix, modulation scheme, and target code rate. However, CSI reports are only useful while they remain representative of the channel at the time of transmission. Increasing the reporting frequency can improve CSI freshness, but also introduces additional uplink feedback overhead and UE processing cost. This project investigates the link-level impact of MIMO configuration, propagation conditions, mobility, and CSI reporting periodicity on downlink throughput.

Methodology

A 5G NR downlink link-level simulation was developed using MATLAB 5G Toolbox. The baseline configuration uses a 3.5 GHz carrier, 40 MHz bandwidth, 30 kHz subcarrier spacing, a 4×4 MIMO antenna configuration, and a CDL-C channel with 300 ns delay spread. CSI feedback is generated using periodic CSI-RS measurements with practical channel estimation. Wideband CQI and PMI reporting, adaptive RI selection, and a Type-I single-panel codebook are used. The main experiments measure successfully delivered PDSCH throughput without HARQ retransmission, allowing the first-transmission impact of CSI-driven link adaptation to be isolated.

The simulation framework is first checked against externally reported fixed-MCS BLER curves. Subsequent experiments examine the baseline link-adaptation behaviour, fixed-rank controls, antenna-configuration and CDL-profile sweeps, and mobility scenarios corresponding to maximum Doppler shifts of 10 Hz, 100 Hz, and 390 Hz. Finally, CSI reporting periods of 4, 10, 64, and 160 slots are compared under each mobility condition.

Key Results and Interpretation

Under the baseline CDL-C 4×4 configuration, throughput increases with SNR and saturates at approximately 164 Mbps. CSI statistics show that the link-adaptation mechanism changes the selected CQI, modulation scheme, code rate, and transmission rank as the channel quality improves. The selected RI increases from one to two layers, but does not exceed two under the baseline channel conditions. Fixed-rank controls confirm that restricting transmission to a single layer limits the high-SNR throughput, while allowing more than two layers provides little additional benefit in this scenario.

Increasing mobility right shifts the throughput curve towards higher SNR. For example, the SNR required to reach 80 Mbps increases from 7.7 dB at 10 Hz to 10.5 dB at 390 Hz. As mobility increases, the channel changes more rapidly, making the reported CSI outdated by the time it is used for transmission. This reduces the accuracy of link-adaptation decisions and leads to lower throughput.

CSI reporting periodicity mainly affects the transition region of the throughput curve rather than the high-SNR saturation level. Relative to the 10-slot baseline, long reporting periods introduce measurable SNR penalties. At 10 Hz, increasing the period to 160 slots produces a 2.4 dB penalty at the selected reference throughput. Reducing the period to 4 slots provides only a modest improvement, with the largest observed gain being approximately 0.5 dB at 100 Hz. Therefore, more frequent CSI reporting is only beneficial under certain mobility and SNR conditions.

Conclusion

The results demonstrate that downlink throughput is jointly determined by channel quality, spatial structure, mobility, and the freshness of CSI feedback. More frequent reporting can reduce the SNR required to achieve a given throughput, but the benefit is scenario-dependent and becomes limited when other effects, such as processing delay or rapid channel variation, dominate. The study provides a link-level basis for future work on adaptive CSI-reporting strategies that explicitly balance downlink gain against uplink overhead and UE power consumption.

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List of Abbreviations

Abbreviation	Definition
3GPP	Third Generation Partnership Project
5G	Fifth Generation
BER	Bit Error Rate
BLER	Block Error Rate
CDL	Clustered Delay Line
CP	Cyclic Prefix
CQI	Channel Quality Indicator
CSI	Channel State Information
CSI-RS	Channel State Information Reference Signal
CW	Codeword
DL	Downlink
DM-RS	Demodulation Reference Signal
FR	Frequency Range
gNB	Next Generation Node B
HARQ	Hybrid Automatic Repeat Request
LDPC	Low-Density Parity-Check
LOS	Line-of-Sight
MCS	Modulation and Coding Scheme
MIMO	Multiple-Input Multiple-Output
NLOS	Non-Line-of-Sight
NR	New Radio
NZP	Non-Zero Power
OFDM	Orthogonal Frequency Division Multiplexing
PDCCH	Physical Downlink Control Channel
PDSCH	Physical Downlink Shared Channel
PMI	Precoding Matrix Indicator
QAM	Quadrature Amplitude Modulation
QPSK	Quadrature Phase Shift Keying
RAN	Radio Access Network
RB	Resource Block
RE	Resource Element
RI	Rank Indicator
SCS	Subcarrier Spacing
SINR	Signal-to-Interference-and-Noise Ratio
SISO	Single-Input Single-Output
SNR	Signal-to-Noise Ratio
TB	Transport Block
TBS	Transport Block Size
TCR	Target Code Rate
TDL	Tapped Delay Line
UE	User Equipment
UL	Uplink
UMa	Urban Macrocell
UMi	Urban Microcell
XPR	Cross polarization power ratios

1 Introduction

1.1 Motivation and Project Scope

Mobile telecommunications have experienced rapid development over the past three decades. Second-generation (2G) mobile systems represented a major milestone in the history of mobile communications, enabling the large-scale adoption of mobile phones for voice and text services. Although 2G was not primarily designed for high-rate data transmission, it established mobile communication as a mass-market technology and introduced the first widely used digital mobile services. Subsequent generations then progressively improved the capability of mobile networks to support packet data transmission. Third-generation (3G) systems enabled more practical mobile internet access, while fourth-generation long-term evolution (4G LTE) in the early 2010s provided substantially higher data rates and lower latency, making services such as video streaming, mobile applications, and real-time online communication widely accessible to end users.

The evolution from 3G to 4G and 5G has been enabled by a combination of physical-layer and system-level innovations. These include the adoption of orthogonal frequency division multiple access (OFDMA), more advanced channel coding and modulation schemes, adaptive modulation and coding, and increasingly sophisticated multi-antenna transmission techniques [1]. In particular, the use of multiple-input multiple-output (MIMO) techniques has become increasingly important as mobile networks have moved towards higher data rates and more reliable wireless links. Rather than relying only on additional bandwidth or higher transmit power, MIMO systems exploit the spatial dimension of the wireless channel by using multiple transmit and receive antennas.

Building upon the capabilities of 4G LTE, 5G New Radio (5G NR) has been designed to support a wide range of service requirements. Three main usage scenarios are commonly associated with 5G: enhanced mobile broadband (eMBB), massive machine-type communications (mMTC), and ultra-reliable and low-latency communications (URLLC) [2, Ch. 1]. These scenarios impose different and sometimes competing demands on the radio access network, including high peak data rates, high spectral efficiency, massive device connectivity, low latency, and high reliability.

MIMO plays a central role in meeting these requirements in 5G NR. By transmitting multiple spatial layers over the same time-frequency resources, spatial multiplexing can significantly increase throughput when the wireless channel supports multiple spatial paths. Beamforming can also be used to direct transmitted energy towards the intended user, improving the received signal quality and reducing interference. In larger antenna-array systems, often referred to as massive MIMO, these benefits can be further enhanced [3]. However, the achievable gains depend strongly on factors such as the antenna configuration, the selected number of layers, the modulation and coding scheme, and the accuracy of the channel information available at the transmitter.

This project focuses specifically on channel-state-information (CSI) feedback for downlink MIMO transmission in 5G NR. In a practical feedback-based downlink system, the base station does not have perfect and instantaneous knowledge of the downlink channel. Instead, the user equipment (UE) estimates the channel using reference signals and reports quantised CSI information back to the base station. This feedback is then used to support link adaptation and MIMO transmission decisions. In 5G NR, the reported CSI may include the channel quality indicator (CQI), rank indicator (RI), and precoding matrix indicator (PMI). These quantities are closely related to the modulation and coding scheme, the number of spatial layers, and the precoder used for downlink data transmission.

The structure and configuration of CSI feedback therefore have a direct impact on system performance. For example, more detailed or more frequent CSI reporting can allow the transmitter to make better-informed decisions, especially when the channel varies rapidly. However, these improvements are not free. CSI-RS and feedback reporting consume downlink and uplink resources, respectively, thereby increasing system overhead and potentially introducing additional processing requirements at both the transmitter and receiver. Moreover, more detailed or more frequent feedback does not always translate into higher throughput, and may instead provide little practical benefit when the additional information cannot be effectively exploited by the system. This creates a trade-off between CSI reporting periodicity, feedback overhead, channel variation, and achievable downlink performance.

The aim of this research project is to investigate how MIMO-related configurations and CSI feedback affect the throughput performance in 5G NR downlink transmission. The study is simulation-based and focuses on the interaction between CSI reporting and link adaptation. In particular, the project examines how antenna configurations, channel models, and CSI reporting periods influence downlink throughput and the overall performance of the system. These parameters are studied under representative channel and mobility conditions in order to characterise the associated trade-offs across different scenarios.

The Third Generation Partnership Project (3GPP) provides the global standardisation framework for mobile communication systems. Originally established for the development of third-generation cellular specifications, its scope has since expanded to cover the continuous evolution of LTE, 5G NR, 5G-Advanced, and future technologies beyond 5G. 5G NR was first introduced through 3GPP Release 15, which defined the first phase of the 5G system and provided the basis for practical 5G deployment. Subsequent releases have continued to enhance the 5G system, with Release 18 and Release 19 forming part of the 5G-Advanced evolution. Future releases are expected to provide the basis for the transition towards 6G [4].

The simulations in this project are conducted in MATLAB using a framework aligned with relevant 3GPP specifications. This alignment includes the use of standardised channel models, such as CDL models, as well as representative baseline system parameters commonly adopted in 5G NR link-level studies, including carrier frequency, bandwidth, subcarrier spacing, antenna configuration, and reference signal structure. The study aims to provide insights that can support future 3GPP standardisation discussions related to MIMO techniques in 5G and 6G.

1.2 Report Structure

The remainder of this report is organised as follows:

- **Section 2: Background and Methodology** introduces the 5G NR downlink physical layer, the overall transmission framework, and the CSI feedback mechanism. It then presents the channel models, Doppler shift, simulation framework, and adopted system configuration.
- **Section 3: Model Validation and Baseline CSI-Feedback Downlink Behaviour** validates the simulation model by comparing the fixed-MCS BLER curve with reported industry reference results. It then presents the baseline CDL-C 4×4 CSI-feedback-driven throughput behaviour and analyses the CSI-driven link adaptation statistics.
- **Section 4: Robustness Across MIMO and Channel Configurations** investigates whether the observed behaviour remains consistent under different spatial and channel conditions. This includes an antenna configuration sweep, a CDL channel profile sweep, and one-layer control experiments.
- **Section 5: Impact of Mobility and CSI Reporting Periodicity** forms the central CSI-focused part of the report. It studies how mobility and CSI reporting periodicity affect CSI-feedback-driven link adaptation, and evaluates whether more frequent CSI updates improve downlink throughput. The section also discusses the practical trade-off between CSI freshness, reporting overhead, and throughput gain.
- **Section 6: Discussion and Conclusion** interprets the main simulation results, discusses the limitations of the study, outlines possible future work, and summarises the implications for 5G NR and future wireless systems.
- **Section 7: References** lists the references used in this report.
- **Section 8: Appendix** provides supporting information.

2 Background and Methodology

This section introduces the technical background and simulation methodology used in this project. The aim is to cover basic related concepts and modelling choices required to study CSI-feedback-driven link adaptation in 5G NR downlink MIMO transmission.

In the first three subsections of this section, an overview of the relevant 5G NR physical layer concepts, including OFDM resource allocation, PDSCH transmission, DM-RS, CSI-RS, and the performance metrics used in this study, are introduced. It then describes the MIMO and CSI feedback framework, such as how the UE estimates the downlink channel and reports CQI, PMI, and RI to support modulation, coding, precoding, and rank adaptation at the transmitter. The CDL channel models and the impact of Doppler shift are also discussed.

The second part of the chapter describes the MATLAB 5G Toolbox simulation framework, the adopted 3GPP-aligned configuration assumptions, and the experimental design.

2.1 5G NR Downlink Physical Layer Overview

The 5G NR downlink physical layer is based on orthogonal frequency division multiplexing (OFDM), where the available time-frequency resources are divided into a two-dimensional resource grid. In the frequency domain, the smallest unit is a subcarrier, while in the time domain the resource grid is divided into OFDM symbols. A resource element (RE) corresponds to one subcarrier in one OFDM symbol. Groups of 12 consecutive subcarriers form a resource block (RB), which is commonly used as the basic frequency-domain allocation unit. In the time domain, transmission is organised into slots. Under the normal cyclic prefix configuration, each slot consists of 14 OFDM symbols, while the slot duration depends on the selected subcarrier spacing. This time-frequency structure provides the basis for allocating data, reference signals, and control-related resources in the downlink [2, Ch. 7].

In this project, the main downlink data channel is the Physical Downlink Shared Channel (PDSCH). The PDSCH carries user data from the base station (gNB) to the UE. Before being mapped to the PDSCH, the transport block is processed by the Downlink Shared Channel (DL-SCH), which includes channel coding and rate matching. The resulting coded bits then undergo modulation and layer mapping. Parameters including the modulation order, target code rate, number of transmission layers, and precoding configuration determine the amount of information that can be transmitted over the allocated resources, as well as the robustness of the transmission. These parameters are central to the link adaptation process studied in this project [2, Ch. 7].

Reference signals are known signals transmitted over selected resource elements, allowing the receiver to estimate or measure the channel for purposes such as data demodulation, CSI feedback, and phase tracking. While there are many types of reference signals, two are particularly relevant to this study: demodulation reference signals (DM-RS) and channel-state-information reference signals (CSI-RS).

DM-RS are associated with the PDSCH and are used by the UE to estimate the effective channel for coherent demodulation of the received data. In contrast, CSI-RS are transmitted to allow the UE to measure the downlink channel and generate CSI reports. Although both are reference signals, they serve different purposes: DM-RS support the decoding of the current PDSCH transmission, while CSI-RS support the transmitter's future link adaptation and MIMO transmission decisions [2, Ch. 9].

The performance of a downlink transmission can be evaluated using several common metrics. The bit error rate (BER) measures the fraction of incorrectly decoded bits, but it is not the most direct metric for the coded transport-block transmission used in 5G NR. In this project, block error rate (BLER) and throughput are more relevant [5].

BLER measures the probability that a transmitted transport block is not decoded successfully. It is especially useful for validating the link-level simulation framework under fixed modulation and coding conditions, where the transmission format is held constant and the decoding reliability can be compared with reference results, as shown in Section 3.1 below.

Throughput is the main performance metric used in this study. It measures the amount of successfully delivered information per unit time, commonly expressed in terms of megabits per second (Mbps), and therefore reflects both reliability and spectral efficiency. A high modulation order or high code rate may increase the potential data rate, but it only improves throughput if the transport blocks can still be decoded successfully. Similarly, transmitting more MIMO layers can increase the peak throughput, but may also increase the risk of decoding failure if the channel cannot support the selected rank.

In practical systems, throughput is also a key metric for network operators because it is closely associated with the user experience.

In practical 5G NR downlink transmission, hybrid automatic repeat request (HARQ) is used to improve reliability. After DL-SCH decoding, the UE sends an acknowledgement or negative acknowledgement (ACK/NACK) to indicate whether the transport block has been decoded successfully. If decoding fails, the gNB may retransmit the transport block using a different redundancy version, allowing the UE to combine multiple transmissions using so-called soft-combining, thereby improving the probability of successful decoding. In this project, HARQ is included in the fixed-MCS validation experiment where comparison with reference BLER results is required, while it is disabled in the main CSI-feedback-driven throughput studies in order to isolate the effect of CSI-based link adaptation [2, Ch. 9] [6].

In summary, the downlink physical-layer components introduced in this section provide the basis for the link-level simulations in this project. PDSCH carries the user data, DM-RS enables coherent demodulation, CSI-RS supports channel measurement for feedback generation, and BLER and throughput are used to evaluate decoding reliability and delivered data rate.

2.2 MIMO Transmission and CSI Feedback

MIMO transmission is a key technique used in 5G NR to improve downlink performance. By using multiple transmit and receive antennas, the system can exploit the spatial dimension of the wireless channel. Depending on the channel condition, antenna configuration and transmission scheme, MIMO can provide array gain, beamforming gain, diversity gain, and spatial multiplexing gain. The main focus of this project is on spatial multiplexing and CSI-driven link adaptation, which use multi-layer transmission to increase throughput.

In spatial multiplexing, the transmitted data stream is divided into one or more spatial layers. The number of layers is often referred to as the transmission rank. If the channel provides sufficiently independent spatial paths, increasing the rank allows more information to be transmitted in parallel, which can significantly increase the achievable throughput. However, using a higher rank is not always beneficial. If the channel is highly correlated, the received signal-to-noise ratio is low, or the receiver cannot reliably separate the layers, then high-rank transmission may lead to a higher block error rate and lower throughput. Rank selection is therefore an important part of MIMO link adaptation.

For a narrowband equivalent MIMO channel, the received signal can be written as

$$\mathbf{y} = \mathbf{H}\mathbf{x} + \mathbf{n},$$

where $\mathbf{H} \in \mathbb{C}^{N_{\mathrm{Rx}} \times N_{\mathrm{Tx}}}$ is the channel matrix, $\mathbf{x}$ is the transmitted signal vector, and $\mathbf{n}$ is additive noise. The number of independent spatial streams that can be supported is related to the rank of $\mathbf{H}$:

$$\mathrm{rank}(\mathbf{H}) \leq \min(N_{\mathrm{Tx}}, N_{\mathrm{Rx}}).$$

where $N_{\mathrm{Tx}}, N_{\mathrm{Rx}}$ are the number of transmit and receive antennas. A channel with a higher rank can support more independent spatial layers, while a low-rank or highly correlated channel limits the benefit of spatial multiplexing.

Precoding defines how the selected transmission layers are mapped onto the physical transmit antennas. The spatial layers are not assigned directly to individual antenna ports. Instead, the layer symbols are multiplied by a precoding matrix before transmission:

$$\mathbf{x} = \mathbf{W}\mathbf{s},$$

where $\mathbf{s}$ contains the transmitted layer symbols, $\mathbf{W}$ is the precoding matrix, and $\mathbf{x}$ is the signal vector transmitted from the antenna ports. For example, in a 4-transmit-antenna system with two layers, matrix $\mathbf{W}$ maps the two layer symbols onto four antenna ports.

The purpose of precoding is to match the transmitted signal more closely to the spatial characteristics of the channel. A suitable precoder can improve the received signal quality and reduce interference between spatial layers, while a poorly matched precoder can make the layers harder to separate at the receiver.

In a feedback-based downlink system, the gNB does not have perfect and instantaneous knowledge of the downlink channel. The UE therefore estimates the channel based on reference signals and feeds back CSI. This feedback allows the gNB to select suitable transmission parameters, including the modulation and coding scheme, the number of layers, and the precoding matrix.

The CSI feedback considered in this project consists mainly of three components: the rank indicator (RI), the precoding matrix indicator (PMI) and the channel quality indicator (CQI).

Among the three components, RI is determined first. The decision of PMI and CQI are selected based on the reported RI. RI reports the number of spatial layers recommended by the UE. This quantity is directly connected to spatial multiplexing. A low RI indicates that the channel is better suited to fewer layers, while a higher RI indicates that the channel can potentially support multiple parallel layers. Since the selected rank affects both the achievable data rate and the reliability of the transmission, RI adaptation is one of the central mechanisms studied in this project. The fixed-rank control experiments in later sections are designed to isolate the effect of this rank adaptation mechanism in some simulations.

PMI specifies the preferred precoder from a predefined codebook. The selected precoder determines how the transmission layers are mapped onto the transmit antenna ports. The precoding matrix indicated by the PMI is selected from a codebook defined by 3GPP TS 38.214, based on the reported RI. A suitable PMI allows the transmitter to better match the spatial structure of the channel, improving the received signal quality and reducing inter-layer interference [7].

CQI represents the estimate of the downlink channel quality made by the UE, which is a scalar 4-bit indicator in the range [0,15]. Under default setting (CQI Table 1), the UE will select the highest possible CQI, so that the downlink BLER does not exceed 0.1 using the selected PMI and under the number of transmission layers defined using RI [8]. A higher CQI corresponds to a more aggressive transmission format, such as a higher modulation order or higher target code rate, while a lower CQI leads to a more robust but lower-rate transmission.

The overall process forms a closed-loop feedback system. The gNB uses the reported CSI to select the transmission parameters, including the modulation scheme, target code rate, and number of layers. The selected configuration is then used for subsequent downlink data transmission and the parameters will be updated when a new CSI report is received.

The accuracy and freshness of CSI feedback are therefore critical to the downlink performance. If the CSI reporting period is too long, the feedback information will be mostly outdated, making the selected transmission parameters no longer suitable. This may cause the BLER to increase rapidly when the channel gets worse or it may lead instead to an overly conservative configuration that fails to exploit the available channel capacity.

CSI reporting periodicity is one of the main parameters controlling this trade-off. A shorter reporting period provides more frequent CSI updates and can improve link adaptation in time-varying channels. However, frequent CSI reporting also increases feedback overhead and may consume additional uplink resources and UE power. The choice of CSI reporting period is therefore not simply a matter of making feedback as frequent as possible, but rather a trade-off between different factors, which will be investigated in detail in Section 5.

A CSI-based downlink transmission and feedback system is summarized in Fig. 1. At the transmitter, the DL-SCH transport block is processed and mapped onto the PDSCH. The PDSCH symbols are then precoded across the transmit antennas and passed through CP-OFDM modulation before being transmitted through the MIMO channel. At the receiver, timing synchronisation, CP-OFDM demodulation, DM-RS-based channel estimation, PDSCH decoding, and DL-SCH decoding are performed.

In parallel with data demodulation, the UE measures CSI-RS resources to estimate the downlink channel for feedback purposes. The resulting CSI report is encoded and fed back to the gNB, where it is decoded and used to select the number of layers, the MCS, and the precoding matrix for subsequent transmissions.

The HARQ feedback path is also presented in Fig. 1. After DL-SCH decoding, the receiver generates an ACK/NACK decision. A failed transport block can be retransmitted and soft-combined with previous transmissions, improving decoding reliability.

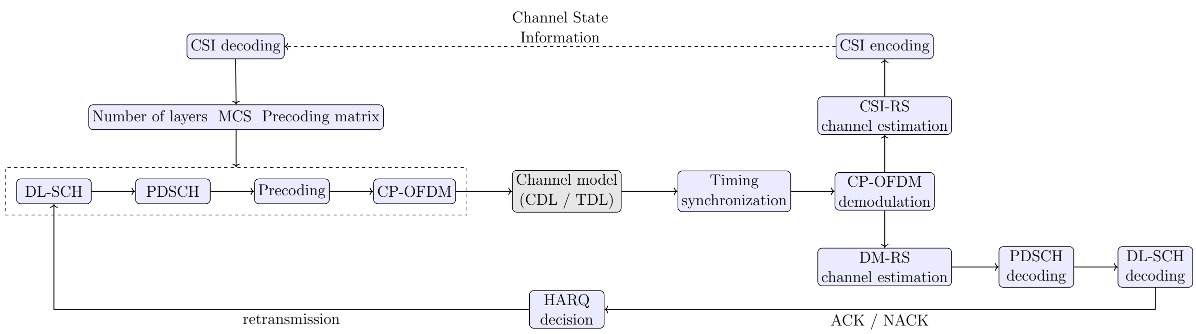

Figure 1: Overview of the CSI-based downlink transmission and feedback framework, adapted from [9]

2.3 CDL Channel Model and Doppler Shift

The performance of MIMO transmission depends strongly on the wireless propagation channel. In a practical mobile environment, the transmitted signal does not usually arrive at the receiver through only a single line-of-sight (LOS) path. Instead, it is reflected, scattered, and diffracted by buildings, vehicles, terrain, and other objects in the environment. The received signal is therefore formed by the superposition of multiple propagation paths with different delays, phases, powers, and angles of arrival.

In order to capture the complex and time-varying nature of the channel, appropriate channel modelling is required in link-level simulation. In 5G NR evaluations, two commonly used classes of channel models are the tapped delay line (TDL) model and the clustered delay line (CDL) model.

The TDL model represents the channel as a set of discrete delay taps, each with an associated path delay and path gain. A simplified time-varying impulse response can be written as

$$h(t,\tau) = \sum_{l=0}^{L-1} h_l(t)\delta(\tau - \tau_l),$$

where $h_l(t)$ is the complex gain of the l-th tap and τ_l is its delay. Because of this relatively compact structure, TDL models are widely used for physical-layer testing and functional validation in 3GPP.

However, a TDL model does not explicitly describe the angular or spatial properties of the propagation paths. This is a limitation for the present study, because the main focus of the simulation is not only to verify whether the downlink transmission chain operates correctly, but also to investigate how systems with different CSI-feedback and MIMO configurations behave under realistic spatial channel conditions. In addition, the potential of MIMO, including multi-layer transmission, cannot be fully evaluated using a channel model lack of spatial features. Similarly, the usefulness of PMI feedback depends on how well the selected precoder matches the spatial structure of the channel.

For this reason, the CDL model is used as the main channel model in this study. Compared with a TDL model, the CDL model provides a more detailed and realistic representation of the MIMO propagation environment. It represents the channel using clusters of multipath components, in which each cluster is associated not only with delay and power but also with angular parameters such as azimuth angle of departure (AoD), azimuth angle of arrival (AoA), zenith angle of departure (ZoD), and zenith angle of arrival (ZoA). The model also includes angular spreads and cross-polarisation properties, allowing it to represent spatial correlation and multi-antenna behaviour more realistically. CDL models are therefore more suitable than simple TDL models for evaluating CSI-feedback-driven MIMO behaviour, because they allow the interaction between channel spatial structure, rank selection, precoding, and throughput to be studied [10].

The CDL channel profiles defined in 3GPP TR 38.901 represent different propagation conditions. CDL-A, CDL-B, and CDL-C are non-line-of-sight (NLOS) profiles, while CDL-D and CDL-E include a line-of-sight (LOS) component. These profiles differ in their cluster delays, relative powers, angular spreads, and cross-polarisation ratios, which leads to different spatial correlation properties and MIMO throughput behaviour [10]. A summary of the CDL channel models defined in 3GPP TR 38.901 is presented in Table 2.

The indicative use cases in Table 2 are related to common urban deployment scenarios. Urban microcell (UMi) scenarios usually represent smaller-cell deployments, where the base station is placed below rooftop level and the link distance is relatively short. Urban macrocell (UMa) scenarios represent larger-cell deployments with a higher base-station height and longer propagation distances.

In this project, CDL-C is selected as the baseline channel model and scaled to an RMS delay spread of 300 ns. This channel setting is commonly used as a representative NLOS baseline in standardisation-oriented link-level evaluations because it provides a rich multipath environment while remaining well defined and reproducible. It is therefore suitable for studying CSI-driven rank adaptation and precoding behaviour without relying on an overly simple channel model. Additional CDL profiles are used in certain experiments to test whether the trends observed under the CDL-C baseline remain consistent under different channel conditions.

Table 2: Summary of CDL channel model profiles based on 3GPP TR 38.901 [10]

Channel Model	**LOS/NLOS**	**Characteristics**	**Indicative Use Case**†
CDL-A	NLOS	Profile with relatively limited angular dispersion compared with CDL-B, and a higher XPR than CDL-B/C.	UMi
CDL-B	NLOS	Profile with the largest azimuth angular spread among the NLOS CDL profiles.	UMi
CDL-C	NLOS	Profile with a richer delayed multipath structure, more clusters, and the smallest departure angular spread among the NLOS CDL profiles.	UMa
CDL-D	LOS	Profile including a specular LOS component.	UMa
CDL-E	LOS	Profile including a specular LOS component, with a longer normalised delay tail than CDL-D.	UMa

† The indicated use cases are interpretative examples rather than fixed definitions in 3GPP TR 38.901. Their association with UMi or UMa scenarios depends on the selected RMS delay spread and the overall simulation setup.

UE mobility leads to time variation of the channel. When the UE moves relative to the base station, the phases of the multipath components change over time. This produces Doppler shifts and causes the channel response to vary overtime. The maximum Doppler shift is given by

$$f_D = \frac{v f_c}{c}, \tag{1}$$

where v is the UE speed, f_c is the carrier frequency, and c is the speed of light, assuming that the base station remains stationary. At a fixed carrier frequency, a higher UE speed corresponds to a larger maximum Doppler shift and a faster time-varying channel. In this study, a maximum Doppler shift of 10 Hz is used as the baseline mobility condition. This corresponds to a low-mobility UE scenario, approximately 3 km/h at a 3.5 GHz carrier frequency, which is close to a typical pedestrian walking speed, and is commonly used as a low-speed condition in 3GPP physical-layer evaluations.

Doppler shift is directly relevant to CSI feedback. CSI reports are generated from channel measurements taken at a particular time, but they are only used by the transmitter after feedback transmission and processing delays. If the channel changes significantly during this interval, the reported CSI available at the gNB may no longer represent the actual channel at the time of PDSCH transmission. This effect is referred to as CSI ageing. Outdated CSI can lead to unsuitable link adaptation decisions, such as choosing an MCS that is too aggressive, selecting more transmission layers than are suitable for the current channel conditions, or applying a precoder that is poorly matched to the instantaneous channel. This is the major focus of this study and is investigated in detail in Section 5.

2.4 MATLAB Simulation Framework

The project is carried out using MATLAB and the 5G Toolbox, with a simulation framework aligned with relevant 3GPP specifications. This alignment includes the use of standardised channel models, such as CDL models, and representative system parameters commonly used in 5G NR link-level studies, including carrier frequency, bandwidth, subcarrier spacing, antenna configuration, and reference signal structure. The use of a 3GPP-aligned simulation framework allows the results to be interpreted in the context of 5G NR physical-layer design, while still providing sufficient flexibility to isolate the effects of individual CSI and MIMO parameters.

The overall MATLAB simulation framework is consistent with the CSI-based downlink transmission and feedback model described in Section 2.2 and shown in Fig. 1. The transmitter performs DL-SCH encoding, PDSCH mapping, MIMO precoding, and OFDM modulation before the waveform is passed through the selected CDL channel. At the receiver, timing synchronisation, OFDM demodulation, channel estimation, PDSCH decoding, and DL-SCH decoding are performed to recover the transmitted transport block. In parallel, CSI-RS measurements are used to generate a CSI report, which is sent to the transmitter and used to select transmission parameters for subsequent transmissions, including the number of layers, the MCS, and the precoding matrix. For the main studies in this project, the HARQ feedback and retransmission loop is not included, so the results reflect first-transmission link performance without retransmission combining.

Table 3 summarises the baseline system and channel configuration adopted in this study. This configuration is aligned with the assumptions commonly used in 3GPP evaluation studies, including TR 38.753 [11]. The full set of parameters used is shown in Table 11 in the Appendix section. Three different mobility scenarios are considered. The low-mobility case assumes a maximum Doppler shift of 10 Hz, corresponding to a UE speed of 3 km/h. For medium- and high-mobility cases, maximum Doppler shifts of 100 Hz and 390 Hz correspond to the speed of approximately 30 km/h and 120 km/h respectively for a 3.5 GHz carrier frequency.

Table 3: System and channel configuration used in the simulations

Parameter	**Value**
Carrier frequency	FR1, 3.5 GHz
Subcarrier spacing	30 kHz
Channel bandwidth	40 MHz (106 RBs)
Antenna configuration	4 Tx, 4 Rx
Maximum number of layers	4
Channel model	CDL-C
Mobility scenarios	10 Hz (Default), 100 Hz, 390 Hz
Simulation type	Downlink link-level

The CDL-C channel model is selected as it represents a typical NLOS urban macro environment and is widely used in 3GPP evaluation studies. A 4×4 MIMO configuration with up to four transmission layers is considered, reflecting a representative frequency range 1 (FR1) deployment while maintaining a manageable simulation complexity. This configuration allows multi-layer transmission as well as rank adaptation to be studied within a single-codeword framework, avoiding additional variability introduced by using a second codeword, and thus enabling a clearer assessment of the effects of the CSI feedback parameters.

The PDSCH, DM-RS, and CSI-RS parameters used are summarised in Table 4. Based on this configuration, a downlink resource grid is shown in Fig. 2 for a CSI-RS-active slot. The first two OFDM symbols are left unallocated because the PDSCH time-domain allocation starts at OFDM symbol index 2. In practice, these initial symbols may contain downlink control signalling, such as PDCCH transmission within a CORESET, although PDCCH is not explicitly modelled in this link-level simulation.

Table 4: Key PDSCH, DM-RS and CSI-RS parameters used for simulation

Configuration	**Parameter**	**Value**
PDSCH configuration	Mapping type	Type A
	Starting symbol	2
	Length	12
PDSCH DM-RS configuration	DM-RS type	Type 1
	Number of additional DM-RS	1
	Max. number of OFDM symbols for DL front-loaded DM-RS	2
	Number of CDM groups without data	1
NZP CSI-RS for CSI acquisition	CSI-RS resource type	Periodic
	Density	1
	Symbol location	4
	Subcarrier location	0

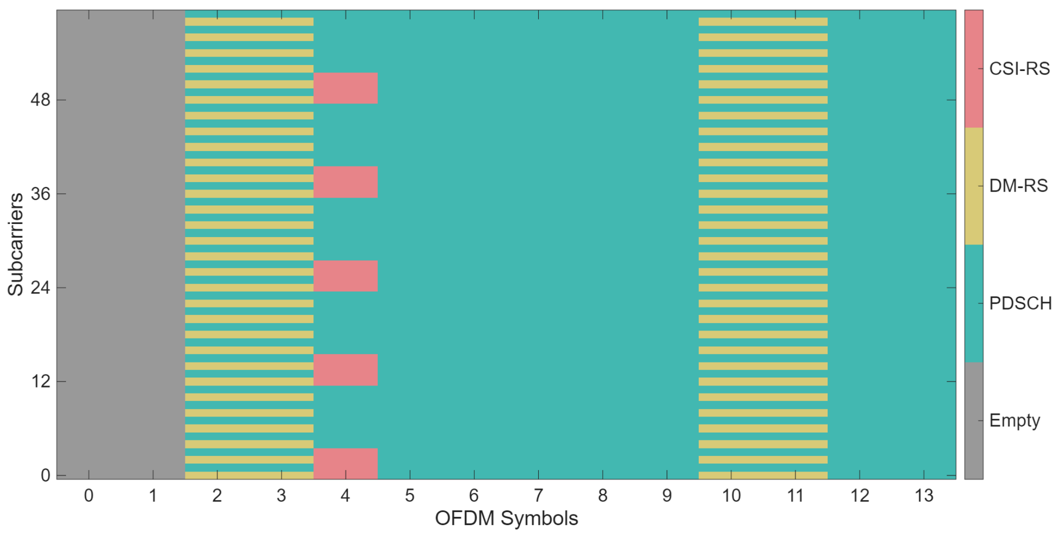

Figure 2: Downlink resource grid showing the allocation of PDSCH, DM-RS and CSI-RS

2.5 System Configuration and Experimental Design

This study focuses on the impact of CSI feedback parameters on downlink transmission performance. The baseline CSI configuration follows the settings specified in relevant 3GPP specifications, including TS 38.101-4 [5], and is summarised in Table 5. Unless otherwise stated, all simulations use the baseline CSI configuration. The full set of CSI configurations used is shown in the Appendix section Table 12.

Table 5: Baseline CSI feedback configuration used in the simulation framework

Parameter	Baseline setting
CSI report periodicity	10 slots
CQI reporting mode	Wideband
PMI reporting mode	Wideband
Subband size	16 RBs
RI configuration	Adaptive
Codebook type	Type I
Codebook mode	Mode 1

In order to ensure a controlled comparison, the simulation studies are designed such that only a limited number of parameters are varied at a time, while the remaining system parameters are fixed at their baseline values. Table 6 provides an overview of the main study groups and their corresponding objectives, together with the specific parameters varied in each group. These studies cover the baseline case, rank adaptation, antenna configuration, CDL channel profile, Doppler shift, and CSI reporting periodicity.

Table 7 then summarises the common execution settings used for all simulation groups, including the SNR sweep, the number of simulated frames per SNR point, and the adopted channel estimation approach. These settings are selected as a trade-off between computational cost, simulation resolution, and the representativeness of the results. For example, the SNR spacing and the number of simulated frames are chosen to avoid unnecessary computational expense, such as using an excessively fine SNR grid or an unnecessarily large number of frames, while still providing sufficient resolution to capture the main performance trends.

Table 6: Summary of simulation study groups and objectives

Study group	Purpose	Varied parameter
Baseline	CDL-C 4×4 reference performance	–
Fixed-rank controls	Layer adaptation	RI $= 1$, RI ≤ 2, RI ≤ 4, RI $= 4$
MIMO configuration sweep	Antenna configuration sensitivity	2×2, 4×4, 8×4
CDL profile sweep	Channel-model robustness	CDL-B, CDL-C, CDL-D
Doppler sweep	Mobility sensitivity	10 Hz, 100 Hz, 390 Hz
CSI period sweep	CSI feedback freshness	4, 10, 64, 160 slots

Table 7: Simulation execution settings

Parameter	Value
SNR range	-5 to 25 dB, with 1 dB spacing
Number of simulated frames	1500 10-ms frames per SNR point
Channel estimation	Practical channel estimation

3 Model Validation and Baseline CSI-Feedback Downlink Behaviour

At the beginning of this section, a simplified scenario is considered in which the adaptation mechanism is temporarily disabled. The modulation scheme, target code rate, and number of transmission layers are fixed, and the simulation results are compared with the results reported by other companies in the 3GPP technical reports. This is used to validate the overall simulation framework.

CSI-based adaptation is then enabled, and the behaviour of the system under the baseline configuration introduced in the previous section is tested. The system performance and the role of CSI feedback are investigated and analysed in detail to provide a clearer understanding of the feedback decisions. In the second half of the section, fixed-rank control experiments are performed to isolate the effects of rank adaptation and multi-layer transmission.

3.1 External Link-Level Validation under Fixed-MCS Transmission

For simulation model validation, an MCS index of 13 (MCS Table 1), 4×4 MIMO, 4-layer, random PMI transmission. A maximum Doppler shift of 10 Hz is used. HARQ is enabled in this simulation. The remaining configuration parameters remain unchanged from the default values defined in Table 3. The modulation order, target code rate, and number of PDSCH layers are fixed, so the results mainly reflect the decoding reliability of a given transmission. The performance is characterised by BLER. The plot is then compared with simulated results reported by different firms, as shown in Fig. 3.

The SNR values corresponding to BLER levels of 70% and 30% are used for comparison in this study. The results are presented in the table below.

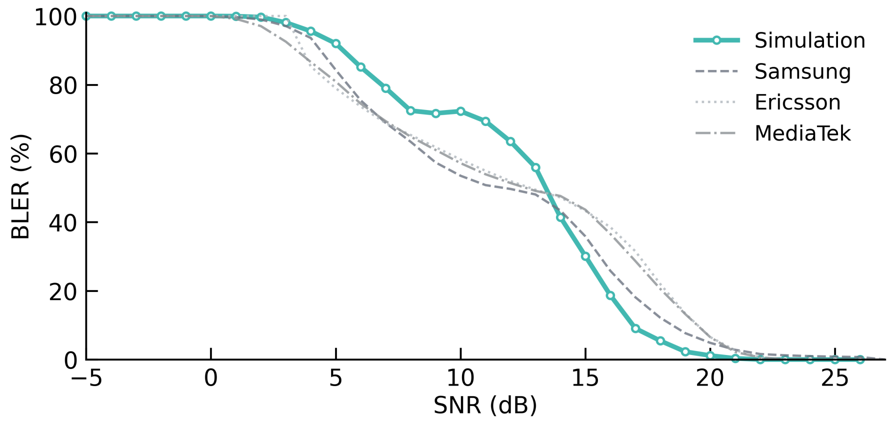

Figure 3: BLER comparison between the simulation and firm-provided baselines, with firm data adapted from [12]

Table 8: Interpolated SNR required to achieve selected BLER levels for the simulation and firm-standard baselines

Target BLER (%)	Simulation	Samsung	Ericsson	MediaTek	Industry Average
70	10.8 dB	6.8 dB	6.8 dB	6.9 dB	6.8 dB
30	15.0 dB	15.6 dB	17.2 dB	16.8 dB	16.5 dB

† SNR values obtained using linear interpolation.

The simulated BLER curve follows the same general trend as the firm-provided baselines: the BLER remains close to 100% at SNR values below 1 dB, decreases as the SNR increases, and approaches zero at SNR values above 23 dB.

In the simulated BLER curve, a plateau-like region is observed at intermediate SNR values, approximately between 7 and 10 dB. This behaviour may be partly attributed to the HARQ mechanism under the fixed-MCS configuration. Around 7 dB, the initial improvement in SNR allows some transport blocks that fail in the first transmission to be recovered through HARQ retransmission and combining, leading to a noticeable reduction in BLER. However, for the remaining more challenging channel realisations, the same fixed modulation, coding rate, and layer configuration may still be too aggressive, so further BLER reduction is limited until the SNR becomes sufficiently high, above approximately 10 dB. Therefore, the plateau-like region can be interpreted as a transition region where HARQ improves the reliability of some blocks, but is not yet sufficient to recover the more difficult decoding cases.

The difference between the simulation and the firm-provided baselines is not unexpected, as the comparison is not based on an identical receiver implementation. At a BLER of 70%, near the plateau-like region, the SNR discrepancy is approximately 4 dB, and at 30% BLER, the discrepancy decreases to approximately 1.5 dB.

Even with the same specified transmission configuration, BLER is sensitive to implementation details such as channel estimation, the LDPC decoding algorithm, iteration count, HARQ combining settings, among other factors. The specific details and the simulation files are not commonly disclosed by the firms. As a result, the reference curves reported by different contributors also differ by approximately 2 dB.

Overall, it can be concluded that the simulated result is in reasonable agreement with the reported reference results. This validation should be interpreted as a qualitative and semi-quantitative alignment of the BLER trend, rather than an exact one-to-one reproduction of a specific BLER curve.

For the subsequent simulation studies, throughput in Mbps is used instead of BLER as the main performance metric. This is because the later experiments involve CSI-driven link adaptation, where the MCS, number of layers and feedback configuration may change across transmission conditions. In such cases, the achievable throughput may every between slots based on the settings, and therefore BLER alone is not sufficient to represent system performance. A configuration may achieve a low BLER simply by selecting a conservative MCS or fewer layers, but this does not necessarily imply high user throughput.

Additionally, throughput is a key performance metric in practical deployments because it directly reflects the useful delivered data rate. Therefore, throughput in Mbps is a more appropriate metric for comparing CSI-feedback and MIMO configurations in the rest of the project.

Furthermore, HARQ was disabled in all subsequent simulation studies in order to isolate the impact of CSI feedback and MIMO-related configurations on the first-transmission performance. This also avoids introducing additional HARQ-related effects, such as the plateau-like behaviour observed in the fixed-MCS validation case discussed above. Since the objective of this project is to compare the relative impact of different CSI configurations rather than to optimise HARQ operation, disabling HARQ provides a cleaner and more controlled basis for comparison.

3.2 Baseline CDL-C 4×4 Throughput Behaviour

Fig. 4 presents the baseline downlink throughput for the CDL-C 4×4 MIMO configuration with adaptive CSI feedback. Over the simulated range of -5 dB to 25dB, the throughput increases with SNR. The curve shows a gradual increase at low SNR, followed by a steeper transition region between approximately 5 dB and 17 dB. At 10 dB, the throughput is approximately 95 Mbps. At high SNR, the throughput saturates at approximately 164 Mbps. The behaviour of the system is explained in the next subsection in terms of the CSI-related decisions. This baseline provides a reference case against which the other scenarios are compared in the subsequent simulations.

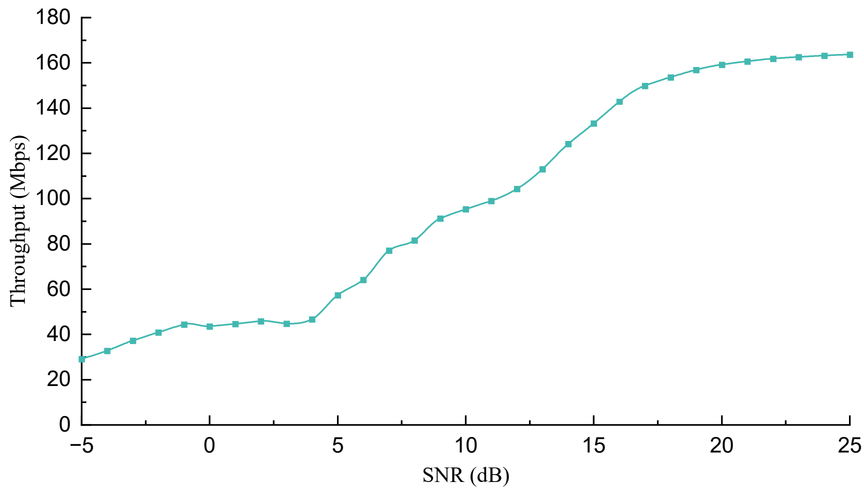

Figure 4: Baseline downlink throughput performance for the CDL-C 4×4 configuration with adaptive CSI feedback

3.3 CSI-driven Link Adaptation Statistics

In addition to the performance of the system, the behaviour of the CSI feedback indicators during the transmission is shown in Fig. 5. The four subfigures respectively show how the CQI, target code rate (TCR), rank indicator (RI), and modulation scheme vary with SNR. For a given SNR point, these quantities are not fixed constants. Instead, they are updated according to the periodic CSI feedback, which depends on the instantaneous channel realisation. Therefore, variations can be observed across slots. Vertical bars are included in the relevant subfigures to characterise the variability of the selected values. The bars indicate the range between the 5th and 95th percentiles of each value rather than the full minimum–maximum range. This choice is used to represent the typical spread of the selected CQI values while reducing the influence of rare deep fades or favourable fading realisations, making the results more representative.

In the CQI plot, as the SNR increases, the selected CQI rises gradually from 5 at low SNR to around 11 at high SNR, indicating that the link adaptation gets progressively more aggressive during the transmission due to the improved channel conditions. Although the mean CQI remains around 11 at high SNR, the highest CQI selected in an individual slot is 14.

Correspondingly, the target code rate of the system follows a similar trend, overall increasing from 0.4 to nearly 0.6. The TCR curve is not strictly increasing but instead having an oscillatory behaviour. The local minima occur at SNR points where major changes in the modulation scheme occur as the system selects a higher-order modulation scheme while temporarily using a lower target code rate. The two major local minima occur at around -3 dB and 4 dB, which correspond to the SNR regions where the dominant modulation scheme changes from QPSK to 16QAM and from 16QAM to 64QAM, respectively.

The selected RI also changes with SNR, reflecting that the system changes the number of layers used for transmission under different channel conditions. At SNR values below 2 dB, the system mainly selects RI = 1, corresponding to robust single-layer transmission. The RI increases to 2 at higher SNR, enabling a steeper increase in throughput. The 4×4 MIMO configuration supports up to four transmission layers in principle. However, the RI does not increase beyond 2 under this scenario.

For MCS table 1 and 3, three modulation schemes are supported, which are QPSK, 16QAM and 64QAM. 256 QAM is only supported when MCS table 2 is used, which is uncommonly used due to its high requirement on channel condition [7]. At very low SNR, the main modulation scheme is QPSK. As the SNR increases, QPSK quickly disappears and 16 QAM becomes dominant. At SNR values above 10 dB, 64QAM modulation dominates. The change in modulation scheme is gradual instead of abrupt, with higher-order modulation schemes being selected more frequently as the SNR increases. At high SNR, the system uses 64 QAM for around 90% of the cases, while 16-QAM is selected in the remaining slots, likely during unusual channel degradation such as deep fades.

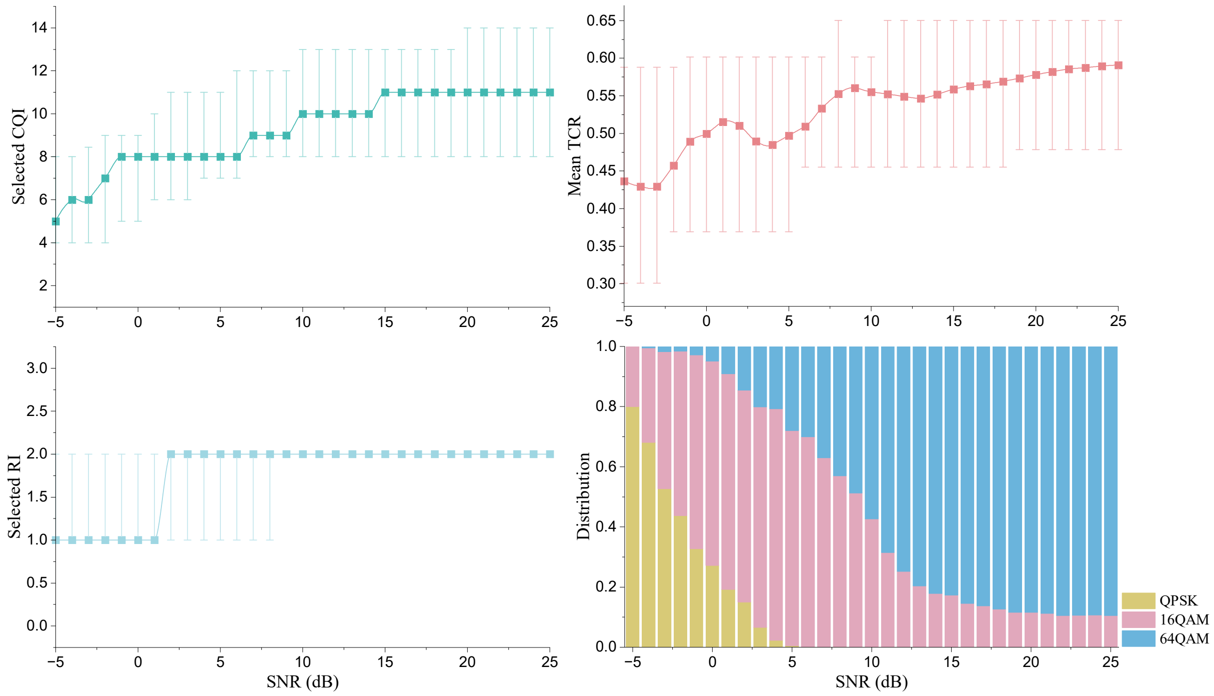

Figure 5: CSI-driven link adaptation behaviour across SNR

The shape of the throughput curve in Fig. 4 can be explained by the CSI-driven link adaptation behaviour shown in Fig. 5.

At very low SNR, the channel conditions are poor and the CSI feedback causes the system to select relatively conservative transmission settings. The selected CQI remains low, leading to the frequent use of QPSK and a low TCR. The selected RI is also mainly one. As a result, the system prioritises link robustness rather than throughput maximisation, and the throughput increases only gradually with SNR.

In the low-to-intermediate SNR range from -1 dB to 4 dB, the throughput curve exhibits a relatively flat region. This behaviour can be interpreted as a transition region of the link adaptation process. Within this region, as the CQI increases, the system begins to select the higher-order modulation scheme, 16-QAM, more frequently, but this is accompanied by a reduction in the target code rate to maintain decoding reliability. Consequently, the gain from using a higher modulation order can be partly offset by the lower coding rate and by the increase in BLER.

The steep increase in throughput in the intermediate SNR range from 4 dB to 18 dB is mainly caused by the combined improvement in modulation, coding rate and spatial multiplexing. As the SNR increases, the selected CQI and target code rate become higher, 64QAM modulation becomes dominant, and the selected RI increases from one to two. The system is therefore able to transmit more bits per resource element and, at the same time, benefit from the additional transmission layer. This combined link adaptation effect leads to a much faster increase in throughput than in the low-SNR region.

At SNR above 20 dB, the throughput gradually saturates. This is because the main link adaptation degrees of freedom have already reached their practical limits under this configuration. In particular, the system is already using 64QAM, which is the highest-order modulation available under MCS table 1. Although there is still theoretical room for a higher target code rate and for RI to increase up to four, the CSI feedback does not tend to select these more aggressive parameters as the SNR further increases. This suggests that the high-SNR performance is no longer primarily limited by the received signal power, but by the practical limits under the selected channel conditions. One possible explanation is that the baseline CDL-C channel does not provide four sufficiently strong and independent spatial layers. Increasing RI beyond two may therefore reduce the effective per-layer reliability and increase the BLER, unless other parameters such as the TCR or modulation order are reduced. As a result, a higher RI does not necessarily lead to higher overall throughput, and the curve naturally saturates around the throughput level supported by two effective spatial layers. The fixed-rank experiments presented in the following subsection are used to examine this interpretation.

3.4 Fixed-Rank Controls

Figure 6 compares the downlink throughput under different rank-restriction configurations. The RI ≤ 4 case corresponds to the original baseline, where up to four transmission layers are permitted. The RI ≤ 2 case restricts the adaptive RI selection to at most two layers, while the RI $= 1$ case forces single-layer transmission.

The RI ≤ 4 and RI ≤ 2 curves are almost identical across the full SNR range. This confirms that, under the configuration considered here, allowing more than two layers provides little additional throughput benefit. This observation is consistent with the link adaptation statistics in Figure 5, where the selected RI increases from one to two at low-to-moderate SNR but does not increase beyond RI $= 2$.

The fixed RI $= 1$ curve behaves differently. At low SNR, single-layer transmission can be slightly more robust than two-layer transmission, leading to lower BER and therefore higher throughput. However, as SNR increases, the lack of spatial multiplexing becomes the dominant limitation. The RI $= 1$ throughput saturates at around 124 Mbps starting from around 17 dB, whereas the adaptive RI cases reach approximately 164 Mbps. This shows that adaptive multi-layer transmission is essential for achieving the high-SNR throughput gain. Higher-rank transmission beyond two layers is not effectively exploited in this baseline scenario. In channels with a higher effective spatial rank and lower inter-layer correlation, adaptive rank selection could provide greater benefits by allowing the system to use even higher ranks during transmission.

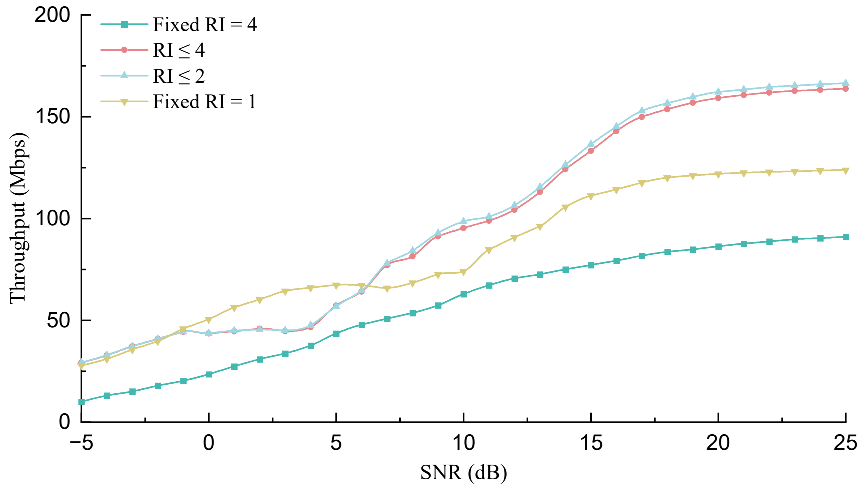

Figure 6: Downlink throughout performance under different maximum layer configurations

3.5 Summary of Baseline Findings

The baseline CDL-C 4×4 simulation provides a reference case for the subsequent studies. The shape of the throughput curve has been investigated in depth, which reflects the discrete and adaptive nature of CSI-driven link adaptation. At low SNR, conservative CQI, coding rate, modulation and RI selections limit the throughput but improve robustness. At the middle SNR range, the throughput increases rapidly as the system moves towards higher CQI values, higher coding rates, higher-order modulation and two-layer transmission. At high SNR, the throughput saturates because the parameters have largely reached their practical limits.

The CSI statistics show that the system adapts in a meaningful way. The selected CQI generally increase with SNR, while the modulation scheme gradually shifts from QPSK to 16QAM and then to 64QAM and TCR increases. The RI selection also changes from single-layer transmission at low SNR to two-layer transmission at higher SNR. However, the system does not select RI values above two in this scenario, suggesting that under the selected CDL-C channel condition, transmission using more than two layers are not well supported.
The fixed-rank control study further confirms this observation. The RI ≤ 2 and RI ≤ 4 cases give almost identical throughput performance, indicating that allowing up to four layers brings little additional benefit in the baseline scenario. In contrast, forcing RI $= 1$ provides a more robust but ultimately throughput-limited transmission mode, especially at high SNR where the lack of spatial multiplexing becomes the dominant bottleneck.
Overall, the baseline results show that the throughput performance is jointly determined by channel quality, CSI feedback, modulation and coding selection, and adaptive rank selection. These findings justify the use of this configuration as the reference baseline for the following studies, where the effects of MIMO configuration, channel profile, mobility and CSI reporting parameters are investigated separately.

4 Robustness across MIMO and Channel Configurations

This chapter examines if the baseline behaviour observed in the previous section remains consistent under different MIMO and CDL channel configurations. The aim of this part is to check whether the main link-adaptation trends remain interpretable when the spatial and channel conditions are changed.

4.1 Antenna Configuration Sweep

Figure 7 compares the downlink throughput performance under different MIMO antenna configurations. The 4×4 case corresponds to the baseline configuration discussed in the previous chapter. Two additional configurations 2×2 and 8×4 are tested.
Despite the fact that, as also shown in the previous section that the maximum number of layers that are actually been used are 2, having a larger antenna array leads to better performance of the system due to .....

The 2×2 configuration leads to a noticeably lower throughput across the SNR ranges as compared to the baseline 4×4 configuration and saturates at around 90 Mbps. This is as expected, as a smaller antenna configuration leads to fewer spatial degrees of freedom.
Oppositely, the 8×4 case can in general lead to better performance of the system. At low-to-medium SNR region, the increase in Tx antennas leads to higher throughput than the baseline, suggesting that the larger transmit array improves the effective channel quality and allows the link adaptation algorithm to select more efficient transmission parameters at lower SNR values ¡?¿. The throughput for baseline outperforms 8×4, however, when the SNR is above 17dB, reaching approximately xxx Mbps compared with around xxx Mbps for 8×4.

This difference should not be interpreted as evidence that 8×4 MIMO is inherently worse at high SNR. The saturated performance of the system is a result of the CSI feedback, which often lead to slightly different CQI, thus different TCR, layer or even transmission schemes. The result here indicates that the additional transmit antennas are not fully converted into a higher peak throughput under the current CSI feedback configuration. In this setup, the benefit of 8×4 is mainly observed as an SNR gain of approximately 5 dB in the low-to-medium SNR region rather than as a higher saturation throughput.

mention what number of layers / RI using rn

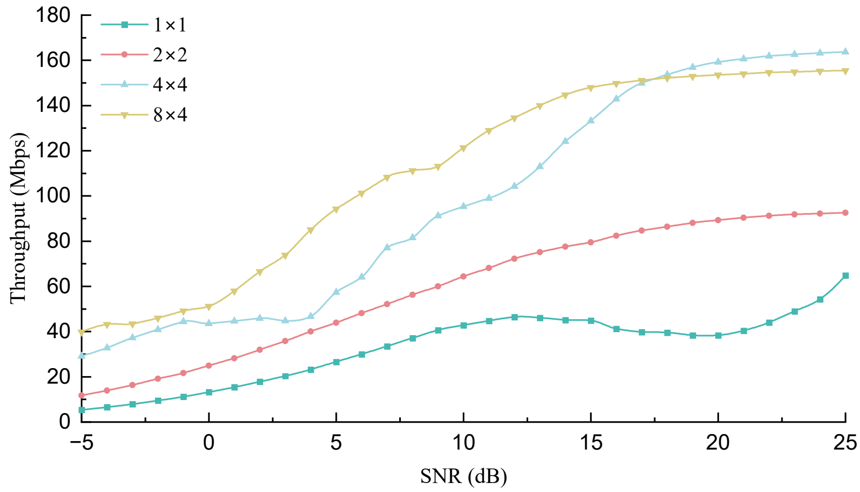

Figure 7: Downlink throughout performance under different MIMO configurations

4.2 CDL Channel Profile Sweep

The second check varies the CDL channel profile while keeping the baseline 4×4 MIMO configuration and CSI feedback settings unchanged.
Figure 8 compares the downlink throughput performance under CDL-B, a typical NLOS channel, and CDL-D, a typical LOS channel with the baseline. The CDL-C case corresponds to the baseline configuration.

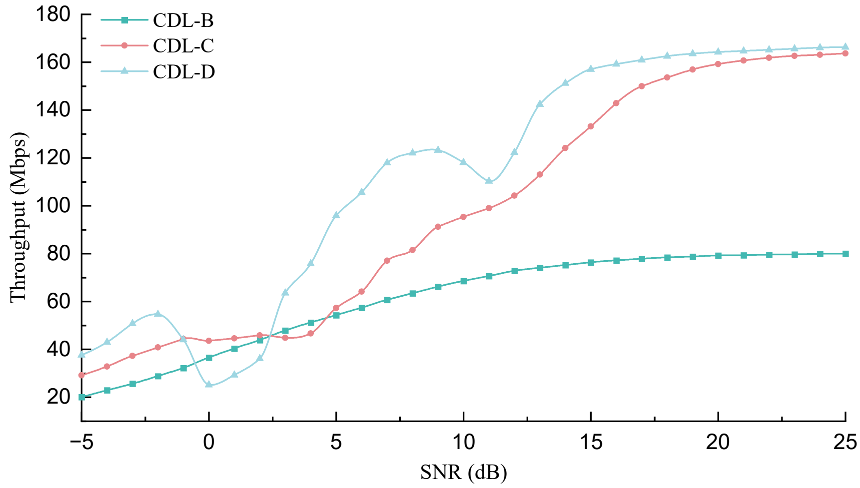

Figure 8: Downlink throughout performance under different CDL channel models

For all three CDL profiles, the throughput increases with SNR and eventually approaches saturated levels, showing that the general link-adaptation behaviour remains consistent.

However, the system performance varies considerably across different channel models. The CDL-B profile increases more gradually, remains below the baseline CDL-C case for most SNR values, and saturates at a much lower throughput of around 80 Mbps. The main difference is found in the modulation and coding selection. Under CDL-B, the CSI feedback tends to report a lower effective channel quality, so the link adaptation procedure selects more conservative MCS values and makes less frequent use of 64QAM. As a result, even when a similar number of spatial layers is used, the lower modulation order and target code rate limit the achievable throughput. This suggests that, although CDL-B has a large azimuth angular spread, the effective post-precoding and post-equalisation SINR in this simulation is less favourable for high-order modulation than in the CDL-C baseline case. The richer delayed multipath structure of CDL-C may provide more exploitable channel diversity under the considered antenna configuration, allowing the CSI feedback to support more aggressive modulation and coding choices.

The CDL-D profile on the other hand achieves the best overall performance. It generally leads to higher throughput at low-to-medium SNR regions and, while leading to almost the same saturated throughput as CDL-C at the high SNR, enters the saturated zone in advance. The overall performance has approximately 5 dB in SNR gain as compared to the baseline. This behaviour is also as expected, as the CDL-D channel has a major LOS component, which supports reliable and rapid transmission even at a low SNR situation, leading to clear advantage as compared to other scenarios, when one-layer transmission is used.

Overall, the CDL profile sweep shows that the general shape of the throughput curve remains largely consistent across different channel models. However, the achievable throughput and the required SNR are strongly channel-dependent. Among the two additional models tested here, CDL-D mainly reduces the SNR required in the transition region, while CDL-B leads to more conservative modulation and coding selections under the considered setup, resulting in a lower high-SNR throughput ceiling.

4.3 One-layer Control Experiments

Similar to section 3.4, a controlled experience is conducted to test the performance of the simulations in this section performs when the maximum number of layers supported is fixed to 1, as shown in Figure 9. In contrast to the adaptive baseline, where the rank indicator is selected through CSI feedback, the transmission rank is constrained to one layer throughout the SNR sweep.
Considering the 4×4 cases, the conclusion from Fig. 8 remains mostly unchanged. CDL-B model is, across the whole simulated SNR range, largely worse than the baseline scenario. CDL-D channel outperforms the baseline at low-and-medium SNR regions, while achieving slightly lower saturated throughput than the baseline.
Compared with the adaptive CDL profile sweep, the fixed one-layer result provides a useful diagnostic reference. In the adaptive case, performance differences may arise from both single-layer link quality and the ability of the channel to support spatial multiplexing. Under fixed one-layer transmission, however, rank adaptation is removed. Therefore, the remaining differences mainly reflect how each CDL profile supports robust single-layer transmission under the same CSI feedback and other configurations.
Considering the sweep of MIMO configurations when the CDL-C channel model is used, the ....

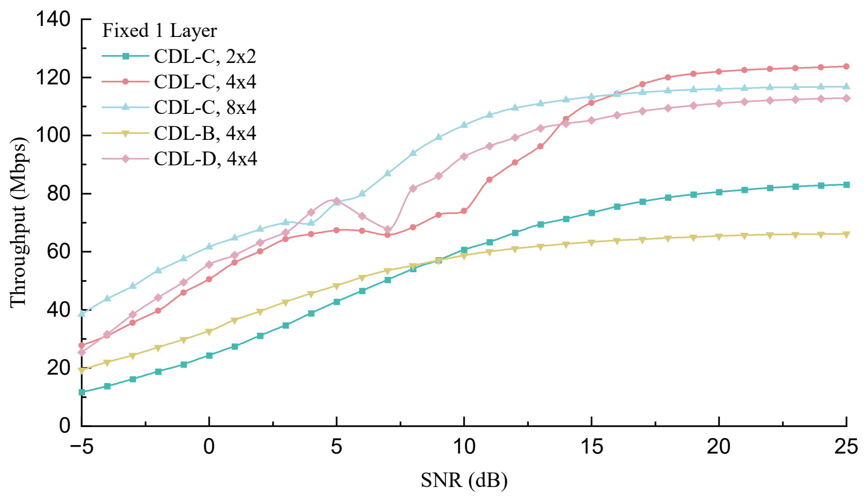

Figure 9: Downlink throughout comparison under fixed one-layer transmission

5 Impact of Mobility and CSI Reporting Periodicity

This chapter investigates the impact of mobility and CSI reporting periodicity on CSI-driven downlink transmission performance. In a time-varying channel, the CSI obtained from CSI-RS measurements may become outdated before it is used to select transmission parameters. This effect is expected to become more significant as the Doppler shift increases or as the CSI reporting period becomes longer. Therefore, this chapter first studies the impact of Doppler shift under the baseline scenario, before examining the effect of different reporting periodicities. Their combined effect is then evaluated to examine the severity of CSI ageing under different mobility conditions.

5.1 Doppler Shift under Default CSI Reporting Period

Figure 10 shows the impact of Doppler shift on downlink throughput under the baseline CSI reporting period. The three Doppler shifts, 10 Hz, 100 Hz, and 390 Hz, represent increasing levels of UE mobility. With a carrier frequency of 3.5 GHz, these correspond approximately to UE speeds of 3 km/h, 30 km/h, and 120 km/h, respectively.

The general throughput behaviour remains similar across all Doppler cases: the throughput increases with SNR, and eventually approaches a high-SNR saturation level. However, higher Doppler shift results in a visible rightward shift of the throughput curve. This indicates that, under higher mobility, a higher SNR is required to achieve the same throughput level.

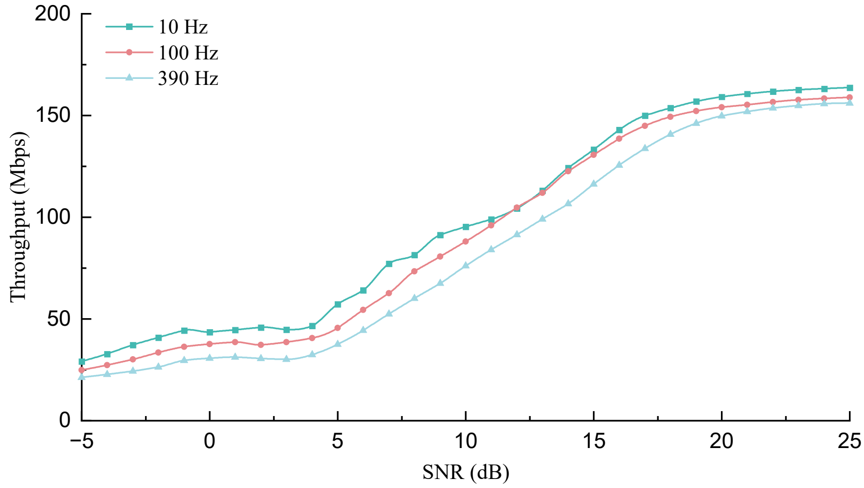

Figure 10: Impact of Doppler shift on downlink throughput

Table 9: Interpolated SNR values at selected target throughput levels

Target Throughput (Mbps)	SNR 10 Hz (dB)	SNR 100 Hz (dB)	SNR 390 Hz (dB)
50	4.3	5.5	6.7
80	7.7	8.9	10.5
110	12.7	12.7	14.4
140	15.7	16.2	17.9

† SNR values obtained using linear interpolation.

This observation is quantified in Table 9, which shows the required SNR for each curve to reach different target throughput levels. The required SNR values are obtained by linear interpolation of the throughput curves. For example, to reach 80 Mbps, the required SNR increases from 7.7 dB at 10 Hz to 8.9 dB at 100 Hz and 10.5 dB at 390 Hz. This corresponds to an SNR penalties of approximately 1.2 dB and 2.8 dB relative to the 10 Hz baseline. Similarly, with a target throughput of 140 Mbps, the 390 Hz case requires 17.9 dB compared with 15.7 dB for the 10 Hz case.

At high SNR, the throughput differences between the mobility cases become smaller, although the higher-Doppler cases remain slightly lower. This suggests that high SNR can partially compensate for Doppler-induced degradation, but cannot completely remove the performance loss caused by faster channel variation.

The degradation is primarily caused by the stronger time variation of the channel under higher Doppler shift. A larger Doppler frequency reduces the channel coherence time, which is approximately inversely proportional to the Doppler frequency, i.e. $T_c \propto 1/f_D$. Therefore, at higher Doppler frequencies, the channel varies more rapidly over consecutive OFDM symbols, slots, and CSI reporting intervals. This makes the received signal more difficult to equalise and decode, since the channel estimated from reference signals may become less representative of the channel experienced by all PDSCH symbols. As a result, this can increase the BLER and reduce the throughput.

In addition to this direct mobility-induced degradation, CSI ageing may further contribute to the performance loss in the closed-loop link adaptation process. Since the CSI reporting period is kept fixed in this experiment, the CSI used by the transmitter becomes increasingly outdated when the channel varies more rapidly. As a result, the selected RI, PMI, and CQI may not match the actual channel conditions at the time of transmission, leading to less effective precoding and less accurate rank and MCS adaptation.

5.2 Reporting Periodicity at Different Mobility

Following the observations in the previous subsection, Fig. 11 compares the effects of CSI reporting periodicity under low- and high-mobility scenarios. In both cases, the CSI-RS and CSI reporting periods are varied together and set to the same value while the remaining configuration parameters are kept at their baseline values.

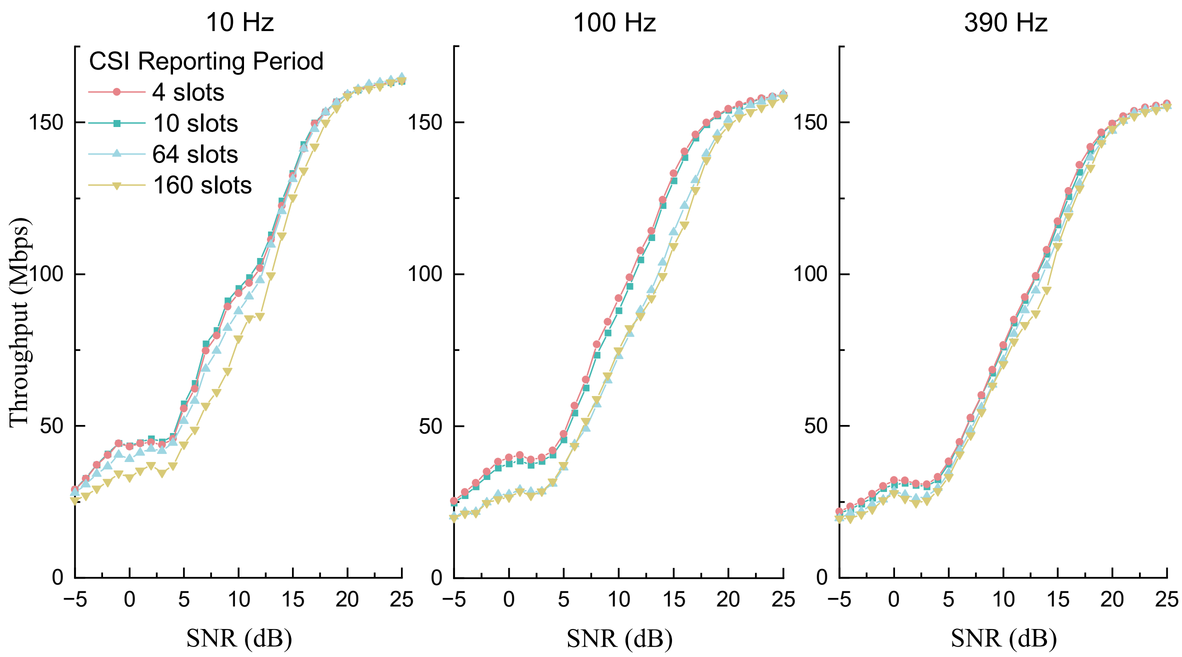

Figure 11: Downlink throughput performance under different CSI reporting periods for low- and high-mobility scenarios

For the 10 Hz case, the throughput curves are relatively close across different reporting periods. At SNR values below 20 dB, the performance of longer reporting periods of 64 slots and 160 slots is clearly perform worse than the baseline reporting period of 10 slots, which is equivalent to a rightward shift of the throughput curve. As the SNR enters the high-SNR region, the performance difference between the reporting periods gradually diminishes. This indicates that the throughput gain from more frequent CSI updates becomes negligible at high SNR. However, reporting every four slots does not generally provide a noticeable improvement compared with reporting every 10 slots.

These results indicate that excessively long reporting periods can degrade performance even under low mobility. However, reducing the period below 10 slots provides limited additional benefit in this scenario.

For the high mobility 390 Hz case, a similar trend is observed. The 4-slot reporting period achieves the best performance in the low-to-intermediate SNR ranges, where longer periods require a higher SNR to reach similar throughput levels. At high SNR, the performance difference becomes small. However, the magnitude of the improvement clearly decreases compared with the low mobility scenario, which is unexpected and will be discussed in the following subsection.

The intermediate Doppler case of 100 Hz shows the most pronounced separation between the CSI reporting periods, particularly between the 4-slot and 160-slot cases. The four reporting periods can be clearly divided into two groups: a higher-performing group with reporting periods of 4 and 10 slots, and a lower-performing group with reporting periods of 64 and 160 slots. This suggests that, under this specific mobility condition, the CSI reporting period should be sufficiently short to track the channel variation effectively. When the reporting interval becomes too long, the CSI used for link adaptation may become outdated, and the performance benefit of frequent CSI updates is significantly reduced. This separation is more pronounced than in the 10 Hz and 390 Hz cases.

Overall, the results show that reducing the CSI reporting period provides a mobility-dependent performance benefit under to the system. The main observed effect is an SNR gain in the transition region rather than a significant increase in high-SNR saturation throughput.

More frequent CSI reporting also involves a trade-off. Although more frequent CSI reporting can improve downlink performance, it also increases uplink feedback overhead, potentially reducing the uplink resources available for data transmission and may increase UE processing load and power consumption. Therefore, more detailed analysis will be conducted to quantify the benefit of having more frequent CSI reporting under different mobility scenarios.

In the following analysis, a CSI reporting period of 10 slots is used as the baseline, denoted as $P_0 = 10$. For each Doppler shift f_D, the reference peak throughput is defined as

$$R_{f_D}^{\max} = \max_{\gamma} R_{f_D,P_0}(\gamma), \tag{2}$$

where $R_{f_D,P}(\gamma)$ is the measured downlink throughput at SNR γ under CSI reporting period P. The target throughput level used for each mobility test case is 50% of the reference peak throughput with 10 slots baseline reporting period.

The SNR required for each reporting period P with fixed mobility f_D to reach the target throughput can be obtained by linear interpolation, denoted as $\gamma_{f_D,P}(0.5R_{f_D}^{\max})$. The change in performance due to the use of different reporting period is evaluated base on the relative SNR penalty or gain with respect to the baseline

$$\Delta\gamma_{f_D,P} = \gamma_{f_D,P}(0.5R_{f_D}^{\max}) - \gamma_{f_D,10}(0.5R_{f_D}^{\max}). \tag{3}$$

Positive values of $\Delta\gamma_{f_D,P}$ indicate an SNR penalty, while negative values indicate an SNR gain. In addition to the slow and fast mobility scenarios tested from Fig. 11, the medium mobility with 100 Hz Doppler shift is also tested. The quantitative results are shown in Table 10 below.

Table 10: SNR required to reach 50% of the baseline reference peak throughput

Doppler	**CSI period**	$\gamma_{50\%}$	$\Delta\gamma$
10 Hz	4 slots	8.2 dB	0.1 dB
10 Hz	10 slots	8.1 dB	–
10 Hz	64 slots	9.0 dB	0.9 dB
10 Hz	160 slots	10.5 dB	2.4 dB
100 Hz	4 slots	8.5 dB	-0.5 dB
100 Hz	10 slots	9.0 dB	–
100 Hz	64 slots	11.0 dB	2.0 dB
100 Hz	160 slots	10.8 dB	1.8 dB
390 Hz	4 slots	10.3 dB	-0.1 dB
390 Hz	10 slots	10.4 dB	–
390 Hz	64 slots	10.8 dB	0.4 dB
390 Hz	160 slots	11.2 dB	0.8 dB

† Reference peak throughput values used: 163.9 Mbps at 10 Hz, 161.0 Mbps at 100 Hz, 158.0 Mbps at 390 Hz.

Table 10 quantifies the relative impact of CSI reporting periodicity within each mobility condition.
At 10 Hz, using very dense CSI reports have only a limited effect. The 4-slot and 10-slot cases require almost the same SNR to reach the target throughput, with only a 0.1 dB difference. However, increasing the reporting period to 64 and 160 slots introduces clear SNR penalties of 0.9 dB and 2.4 dB, respectively. This indicates that although the channel is slowly varying at 10 Hz, very infrequent CSI reporting can still degrade the performance at mid SNR region because the feedback becomes less representative of the current channel state.

For the 390 Hz case, consistent with the observation from Fig. 11, having lower reporting period leads to penalties of 0.4 dB and 0.8 dB relative to the baseline, which is smaller than the penalty for 10 Hz scenario, which are 0.9 dB and 2.4 dB respectively. Increasing the report frequency to every four slots also only leads to ignorable amount of improvements.

The impact of CSI period change has more visible impact for the 100 Hz scenario. The 4-slot case achieves a 0.5 dB SNR gain relative to the 10-slot baseline, showing that more frequent reporting does lead to better performance. In addition, slowing down the report to every 64-slot and 160-slot causes 2.0 dB and 1.8 dB additional SNR, respectively, much more dominant than the other two mobility scenarios.

The non-monotonic difference between 64 and 160 slots for the 100 Hz case and between 4 and 10 slots for the 10 Hz case should not be over-interpreted, as it is likely due to simulation inaccuracy, limited number of frames simulated, and the discrete nature of CSI-driven CQI, PMI, and RI decisions.
Overall, the results from this subsection show that CSI reporting periodicity mainly affects the SNR required in the transition region rather than the final saturation throughput. Shorter reporting periods can provide an SNR gain by reducing CSI ageing, especially under moderate mobility. Under low mobility (3km/h) case, having denser CSI period has only limited benefit to the transmission as the channel is not undergo rapid change. Under high mobility (120km/h) case, similar observation is observed, but the underlying reason is completely the opposite, since the channel is changing too-rapidly. Having short reporting period alone in this scenario does not fully recover the loss caused by rapid channel variation. The CSI is affected not only by the reporting period, but also by factors such as processing delay, channel estimation accuracy, and the time variation between CSI report and transmission, where these issues become especially dominant under high mobility.

5.3 Discussion of CSI Freshness and Practical Trade-Off

The results in Sections 5.1 and 5.2 show that the value of CSI feedback should not be evaluated only by its reporting granularity or reporting frequency. Instead, its effectiveness depends on the environment, especially the mobility of the system that impacts the channel directly. A visualization of the relative SNR offset under different situation is shown in Fig. 12 below based on data from Table 10.

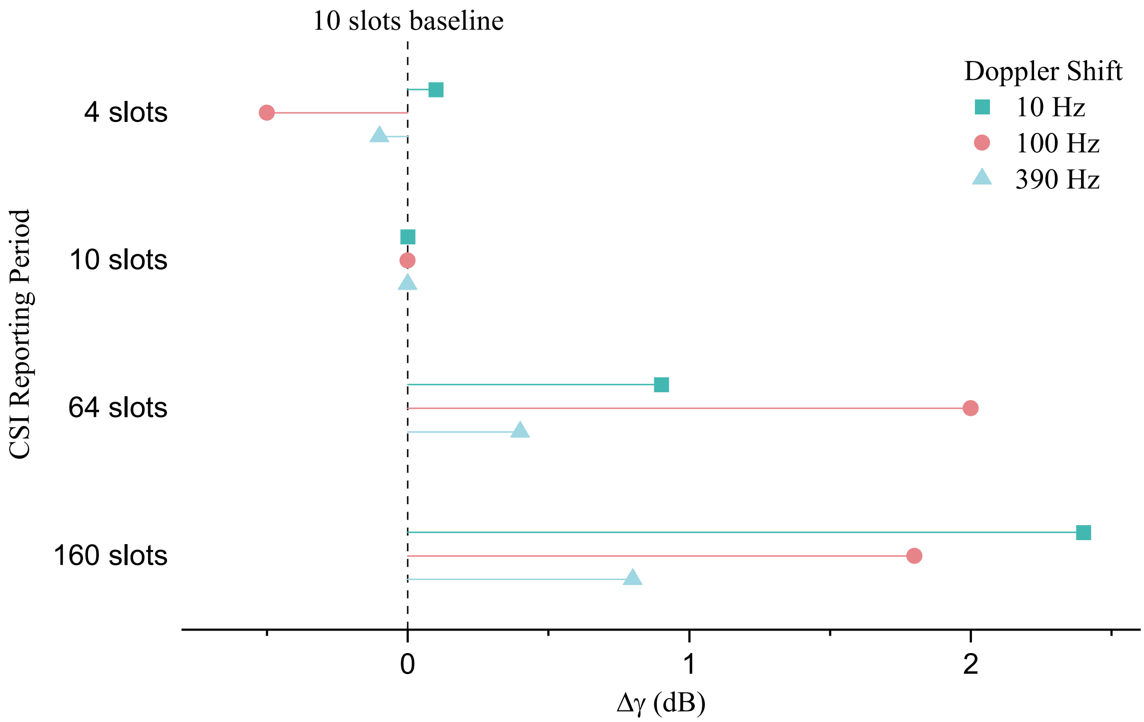

Figure 12: Relative SNR offset $\Delta\gamma$ with respect to the 10-slot CSI reporting baseline

Based on the results, it is clear that a CSI reporting period is suitable to be used as the default setting during the transmission. It in general provides a satisfactory level of performance for the system while avoided the use of overly-frequent CSI reports. The 10 slots period is especially suitable as, for the 10 Hz and 100 Hz scenario, significant performance degradation will occur when the CSI period decreases from per 10 slots to 64 slots (0.9 dB and 2 dB respectively).

However, the results also shows that the system has the potential to adjust the CSI periodicity based on the channel condition, thus achieving a trade-off balance between the performance and the uplink throught as well as the UE power consumption. For example, when the UE is at medium-speed around 30 km/h, it has been proven that the system can enjoy a noticeable improvement in the throughput when increasing the reporting period from default baseline to per 4 slots. In the meantime, for high-speed scenarios such as on a train or motorway, using the default value may even be considered as a waste of energy for the UE. a longer reporting period around 64 slots, may be more suitable. Intermediate values, such as 32 or 40 slots, may also provide a better trade-off, although these configurations have not yet been evaluated. In addition, Fig. 11 has also showed that at high SNR, the throughput curves for different reporting periods converge to similar saturation levels. This suggests that having frequent CSI report is only beneficial at low-to-medium SNR regions. At high SNR regions, using per 64 slots or 160 slots reporting is already enough for the system.

More frequent CSI reporting improves the freshness of CSI and can reduce the SNR required to achieve a given throughput, but it also increases uplink feedback overhead and consumes more power of the UE. This overhead includes the resources required to transmit RI, PMI, and CQI reports, and becomes more significant when reporting is frequent. The observation here provides an insight to the CSI reporting period adjustment algorithm in terms of finding the most suitable reporting period based on the environment and channel status.

Moreover, the simulation results suggest that the optimal CSI reporting period is mobility- and scenario-dependent. In addition to the fixed rules discussed in the previous paragraphs, a closed-loop feedback algorithm could be introduced during transmission to adaptively select the CSI reporting period. Starting from a default reporting period, or from an empirically selected value based on the current scenario, the system could adjust the reporting period among neighbouring options and measure the resulting gain or penalty. If the improvement is considered significant enough to justify the additional feedback overhead, the new CSI reporting period could be adopted for a predefined duration before being re-evaluated.

When looking further towards future AI-native 6G systems, this type of adaptation could also be supported by learning-based control. A related example is given in [13], where deep neural network (DNN) and reinforcement-learning (RL) based schemes are used to dynamically adjust the CSI feedback interval by balancing channel estimation accuracy against feedback overhead. This suggests that AI-native control could be a promising extension for adaptive CSI reporting periodicity, although practical deployment would still need to consider factors such as processing delay, complexity, stability and standardisation support.

6 Discussion and Conclusion

6.1 Interpretation of Results

The results from the previous sections confirm that CSI feedback plays an important role in link adaptation in the simulated 5G NR downlink system.

By allowing the transmitter to optimise rank, modulation scheme, target code rate, and PMI according to the estimated channel conditions, the system can avoid some inefficient fixed-transmission choices and therefore achieve better performance under the given channel conditions. The benefit of rank adaptation is particularly clear from the fixed-rank control experiments in Section 3.4, where constraining the rank leads to performance loss when the channel would otherwise support a more suitable number of transmission layers.

The robustness experiments presented in Section 4 further show that the system exhibits broadly consistent behaviour across different simulation settings. The results indicate that different channel models and antenna configurations produce similar throughput trends, although the absolute throughput values can vary depending on the channel characteristics and the array or multiplexing gain provided by the MIMO configuration.

In Section 5, an in-depth investigation into the selection of the CSI reporting period based on channel conditions has been conducted. The results show that the benefit of fresher CSI must be balanced against the additional reporting overhead. More frequent

CSI reports can noticeably improve the accuracy of link adaptation when the channel varies sufficiently fast and when the system operates in a low-to-intermediate SNR region where the throughput curve is sensitive to CSI quality. However, once the system approaches the high-SNR saturation region, or when either the UE is nearly stationary or moving very fast, the extra feedback overhead may provide only limited downlink benefit, while potentially increasing UE power consumption and reducing the uplink resources available for data transmission. A closed-loop feedback algorithm for adaptive reporting period selection has also been proposed to balance downlink performance against feedback overheads.

Overall, the project shows that CSI feedback can contribute significantly to the performance of the downlink transmission. The reporting configuration should be selected depending on the operating scenario, rather than being fixed purely by a conservative default value or simply targeting maximum possible downlink throughput.

6.2 Limitations

The study is based on link-level simulation. It therefore focuses on the behaviour of a single downlink connection under controlled channel and link-adaptation settings. Higher-layer and system-level effects, such as multi-user scheduling, MU-MIMO operation, handover, and cell-level resource allocation, are not explicitly modelled. In addition, although CSI reporting is considered in terms of its feedback periodicity, the uplink physical channel is not separately simulated. As a result, the conclusions are most directly applicable to physical-layer link adaptation rather than complete system-level network optimisation. Since only downlink is modelled, the quantitative cost of CSI reporting is not directly evaluated. How the feedback overhead will lead to additional uplink resource consumption, which may affect the uplink throughput and UE power consumption, is not investigated. The trade-off between downlink throughput improvement and uplink reporting cost is therefore evaluated mainly from the observed downlink performance trends.

Furthermore, HARQ is disabled in the main simulation studies to simplify the analysis and isolate the first-transmission performance of different CSI and MIMO configurations. This simplification provides a cleaner basis for comparing the relative trends between different setups. However, the absolute throughput in a practical system may differ from the simulated results, since HARQ retransmissions and soft combining can improve decoding reliability, especially in the low-to-intermediate SNR regions.

The simulation uses a selected set of carrier, antenna, channel, Doppler, and CSI-reporting configurations based on the default configuration commonly set during 3GPP requirement tests. These choices are sufficient to reveal the main trends, but they do not cover the full 5G NR parameter space. Many other parameters, such as codebook types and wideband and subband CQI reporting configurations, may lead to different quantitative results.

Finally, although practical factors such as the UE processing delay for CSI reporting have been considered in the simulation, the receiver and channel-estimation assumptions remain simplified compared with a practical implementation. Practical hardware impairments, calibration errors, imperfect synchronisation, and other implementation-specific effects are not fully represented. Therefore, the absolute throughput values should be interpreted as simulation-based indicators rather than direct predictions of field performance.

6.3 Future Work

Future work could first introduce an explicit feedback-overhead model to quantify the uplink cost associated with different CSI reporting periods. The proposed adaptive CSI-reporting algorithm could then be implemented and tested. This would turn the empirical observations in this project into a practical feedback-control strategy that can be evaluated as a practical feedback-control strategy.

Further work could also extend the simulation to a wider range of deployment scenarios, including different channel conditions and antenna configurations. A system-level study that incorporates MU-MIMO would be useful for evaluating whether the link-level conclusions remain valid when resources are shared among multiple users. The adaptation algorithm is also expected to become more complex when the base station serves multiple UEs

Finally, due to the limited time available for this project, only a limited set of CSI settings has been investigated. Future work could explore more advanced CSI-adaptive strategies by jointly considering different CSI configurations under more diverse transmission conditions. Such an extended simulation framework could also provide a foundation for AI-assisted adaptation in future 6G systems, in which ML-based algorithms can be trained and tested.

6.4 Implications for 5G NR and Future Wireless Systems

The findings of this project have broader implications beyond the specific simulation cases considered in this report. Although the study is based on link-level simulation and cannot be seen as a direct prediction of system performance in real-life testing conditions, it provides useful insights for standardisation-oriented discussions, for example for 3GPP, on the design and refinement of CSI reporting mechanisms.

In practical system design, the objective is not only to maximise downlink throughput, but also to balance downlink performance against uplink performance, UE power consumption, implementation complexity, compatibility, and robustness under a range of different conditions. Accordingly, the simulated results suggest that overly frequent CSI reporting may not always be justified, especially when the resulting downlink throughput gain is limited or when the system already operates near the high-SNR saturation region. Therefore, an adaptive CSI reporting strategy, such as the one proposed in this study, may provide a more efficient solution than using a fixed configuration across all scenarios.

The simulation framework developed in this project is based on 5G NR. Although 6G standardisation is still at an early stage, future systems are expected to retain the need for channel-state feedback and link adaptation. The trade-off between CSI freshness and reporting overhead may become increasingly important in future wireless systems, including 6G. While many existing mobile services such as video streaming and content delivery are downlink-intensive, future applications may create stronger uplink demand, including high-resolution live streaming, extended-reality interactions, and high volume data exchange. In such scenarios, uplink performance may become increasingly important alongside downlink performance. In such scenarios, uplink resources may become increasingly valuable, and excessive CSI feedback may compete more frequently with user-data transmission. This further strengthens the need to evaluate CSI reporting and other related parameters.

Overall, this project supports a broader design direction in which CSI feedback configuration should be selected according to scenario-dependent system utility. Rather than configuring CSI reporting solely to maximise peak downlink throughput, future 5G and 6G systems could benefit from adaptive mechanisms that jointly consider channel variation, mobility, SNR region, and other factors such as the uplink throughput and power consumption. This would allow CSI reporting to contribute to more efficient, robust, and balanced wireless communication systems.

7 References

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8 Appendix

8.1 Risk Assessment

A complete risk assessment was provided in the submitted Technical Milestone Report (TMR). Since this project is entirely simulation-based and does not involve any physical laboratory experiments, no safety-related or hardware-related risks were identified. The only practical risk identified was the limited time available for the project relative to the computational cost of the simulations.

The risk assessment presented in the TMR remains valid, and the identified computational risk has been carefully managed throughout the project. For example, during the development of the simulation program, particular care was taken to ensure that the code could be executed in parallel on multiple CPU workers using the MATLAB Parallel Computing Toolbox, rather than being limited to a single CPU thread at a time. In addition, external computing resources were used where necessary, including the CUED central computing system `ts-access`.

8.2 Ethical Statement

This project is entirely simulation-based and does not involve personal data or physical laboratory experiments. Therefore, no direct safety, privacy, or human-related ethical risks are identified.

From an environmental perspective, the project requires computational resources for MATLAB simulations. Although the total computational demand is very limited relative to most other simulation-based research projects, unnecessary computation was avoided where possible by using a consistent baseline setup, a clearly pre-defined experimental structure and an appropriate number of simulated frames to balance result accuracy and computational cost.

Additionally, the findings of the project may have positive social and environmental implications, as they provide a detailed analysis of CSI feedback and MIMO techniques, which could potentially contribute to more efficient wireless communication system. For example, the results may help avoid using overly frequent uplink reporting when it is unnecessary. Improved link adaptation can also support better spectral efficiency and more reliable connectivity.

8.3 Declaration of the use of generative AI

Generative AI tools, specifically ChatGPT and Google Gemini, were used in a limited capacity during the completion of the project. AI assistance was used mainly during the early stage of the project, including to facilitate the understanding of related concepts and to support the debugging of the MATLAB simulation implementation.

It can be confirmed that the experimental design, data analysis, interpretation of results, final conclusions, and report content were completed and verified by the author. No AI tool was used, in any form, to generate, modify, or manipulate the simulated data.

8.4 Detailed Baseline Simulation Configuration

Table 11: Baseline simulation parameters, adapted from TR 38.753 where applicable [11]

Parameter		Value
Duplex mode		TDD
FR / Carrier frequency		FR1, 3.5GHz
Maximum Doppler shift		10 Hz
Channel Bandwidth / SCS		40Mhz / 30kHz
Maximum allowed rank		4
Antenna configuration		4Tx 4Rx
Channel model		CDL-C
Delay spread		300 ns
Codebook configuration for PDSCH and DMRS		Single Panel Type I
MCS table		Table 1 (64 QAM)
PDSCH configuration	Mapping type	Type A
	k0	0
	Starting symbol	2
	Length	12
PDSCH DMRS configuration	DMRS Type	Type 1
	Number of additional DMRS	1
	Max. number of OFDM symbols for DL front loaded DMRS	2

Table 12: Baseline CSI simulation parameters, adapted from TR 38.753 where applicable [11]

Parameter		Unit	Value
NZP CSI-RS for CSI acquisition	CSI-RS resource type		Periodic
	Density		1
	Symbol location		4
	Subcarrier location		0
	CSI-RS interval and offset	slot	10, 1
CQI			Wideband
PMI			Wideband
CQI/RI/PMI delay		ms	7

8.5 Project Repository and Supporting Materials

The simulation code and supporting materials developed for this project have been made available in a GitHub repository: `https://github.com/Bariumoxid/MIMOTechniques`. The repository includes the main MATLAB simulation scripts, configuration files, workflows, and representative test cases used to generate and validate the results presented in this report. It also contains additional supporting materials, including weekly progress reports, the logbook, and source files for the report and figures.